Ping Wang

pingw220@gmail.com | github.com/pingw220

Research Interests

Human-Aligned Generative Models, Preference-Based Post-Training, Multimodal Learning and Evaluation, Reinforcement Learning, Music and Audio Generation, and Music Information Retrieval.

Education

University of Washington Seattle, WA
B.S., Paul G. Allen School of Computer Science & Engineering; GPA: 3.91/4.0 *Jun 2027*

- **Relevant Coursework**: Machine Learning, Natural Language Processing, Data Structures and Algorithms, Distributed Systems, Operating Systems, and Computer Networks.

Publications and Preprints

Ping Wang, Guang Yang, Shao-Rong Su, Junkai Wu, Pang Wei Koh, and Noah A. Smith. *Rubric-Based Optimization for Text-to-Music Generation*. Under review at ICLR 2027. [arXiv].

Ping Wang, Guang Yang, Nazif Can Tamer, Victoria Ebert, and Noah A. Smith. *A Dual Evaluation for Music Transcription*. Under review at TISMIR. [arXiv].

Research Experience

Music and AI Lab, National Taiwan University Taipei, Taiwan
Student Researcher — Advisor: Prof. Yi-Hsuan Yang *May 2026 – Present*

- **SheetSage-Pro**: Built a bilingual, lyric-aligned audio-to-lead-sheet transcription pipeline for extracting melody, chords, beats, meter, and musical structure from full-song audio. Evaluated beat tracking, pitch correction, and rhythm quantization to improve transcription quality. Accepted for presentation at ISMIR 2026 Late-Breaking/Demo.
- **Lyrics-to-Song Generation with MIDI-LLM and MIDI-SAG**: Developing an end-to-end lyrics-conditioned song generation pipeline by connecting symbolic lead-sheet generation with downstream arrangement and audio synthesis. Fine-tuning language models to generate melody, chords, rhythm, and lyric-aligned symbolic representations, including syllable-level lyric pointers, phonetic information, rests, breaths, and many-to-many lyric–note alignment for controllable generation.

Noah's Ark Lab, University of Washington Seattle, WA
Student Researcher — Advisor: Prof. Noah A. Smith *Aug 2025 – Present*

- **A Dual Evaluation for Music Transcription**: Developed a dual evaluation framework for assessing both notation accuracy and performance fidelity in automatic music transcription. Evaluated 24 transcription pipelines composed of eight audio-to-MIDI models and three MIDI-to-score systems across 230 piano performances. Designed and analyzed an audio-only human listening study with 106 participants and 3,180 pairwise judgments, and compared human preferences against notation-based, signal-processing, and learned audio similarity metrics.
- **Rubric-Based Optimization for Text-to-Music Generation**: Developed structured, audio-language-model rubric rewards for text-to-music post-training. Constructed preference pairs for direct preference optimization (DPO) on MusicGen-small and ACE-Step v1 and applied scalar rewards with DiffusionNFT on ACE-Step. Evaluated quality and controllability with independent perceptual and objective metrics; found broad gains from rubric rewards and advantages of specialized rewards for precise musical attributes.

UbiComp Lab, University of Washington Seattle, WA
Student Researcher *Sep 2024 – Oct 2025*

- **Embedded Gesture Recognition**: Designed and optimized lightweight neural networks for real-time gesture recognition on resource-constrained embedded devices, including model quantization and deployment for on-device inference.
- **Tuberculosis Screening from Cough Audio**: Applied transformer-based architectures, including Audio Spectrogram Transformer (AST) and MobileViT, to cough-based tuberculosis risk classification.
- **Cough Detection and Localization**: Developed machine learning models for cough-event detection and temporal localization in continuous audio recordings.

Industry Experience

Amazon Taipei, Taiwan
Software Development Engineer Intern *Jun 2025 – Sep 2025*

- **Embedded Firmware Development**: Designed and implemented embedded firmware for inter-chip communication, camera-module integration, and Wi-Fi connectivity validation for smart-home devices.
- **Hardware–Software Integration**: Worked on board-level integration and debugging across embedded hardware, firmware, camera modules, and network interfaces.
- **Testing Infrastructure**: Developed automated unit tests and peripheral mock frameworks for validating firmware across hardware states, improving test coverage and reducing regression risk.

Phison Electronics Miaoli, Taiwan
Machine Learning / Embedded Systems Engineer Intern *Jun 2024 – Aug 2024*

- **LLM Memory Modeling**: Developed a parameter-calculation system for estimating large language model memory consumption, including model weights and activations, directly from model configuration files.
- **Transformer Architecture Analysis**: Analyzed LLaMA 2/3, CodeLlama, Falcon, Mistral, Mixtral, Breeze, Whisper, and Vicuna, with emphasis on attention architectures, model scaling, and memory requirements.
- **Large-Scale Model Training**: Worked with multi-GPU Linux infrastructure and Phison AI SSD systems for model training and experimentation.

Technical Skills

Programming Languages: Python, C, C++, Java
Machine Learning: PyTorch, Transformers, Preference Optimization (DPO), Reinforcement Learning, Diffusion / Flow-Matching Models, LoRA
Music and Audio AI: Automatic Music Transcription, Music Generation, Audio Representation Learning, Symbolic Music Modeling, MIDI, MusicXML
Systems and Tools: Linux, Git, Distributed Training, Multi-GPU Computing, Embedded Systems, FPGA